A NOETHERIAN SPACE IS QUASI-COMPACT, AND SO IS EVERY SUBSET

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The statement that every subset of a Noetherian space is Noetherian. Invoked at the end of the counted proof (source lines 2166-2168). Deduces the second assertion from the first.

Source reference. Original lemma lemma-Noetherian: sources/stacks-topology/pinned-original-source.tex, source lines 1470–1470. Navigation pointer only: the counted text refers to the Stacks item lemma-Noetherian, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let X be a Noetherian topological space.*

(1) *The space X is quasi-compact.*
(2) *Any subset of X is retrocompact.*

Proof. Suppose $X = \bigcup U_i$ is an open covering of X indexed by the set I which does not have a refinement by a finite open covering. Choose $i_1, i_2, \ldots$ elements of I inductively in the following way: Choose i_{n+1} such that $U_{i_{n+1}}$ is not contained in $U_{i_1} \cup \ldots \cup U_{i_n}$. Thus we see that $X \supset (X \setminus U_{i_1}) \supset (X \setminus U_{i_1} \cup U_{i_2}) \supset \ldots$ is a strictly decreasing infinite sequence of closed subsets. This contradicts the fact that X is Noetherian. This proves the first assertion. The second assertion is now clear since every subset of X is Noetherian by Lemma lemma-Noetherian. □

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